

JAPNIT SINGH SAWHNEY

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EDUCATION

Thapar Institute Of Engineering & Technology, Patiala	Aug 2023 - Present
B.Tech in Computer Engineering	CGPA: 9.10
St. Mary's Academy, Meerut	Apr 2022 - Mar 2023
Senior Secondary School certificate	Percentage: 94.5%
St. Mary's Academy, Meerut	Apr 2020 - Mar 2021
Senior Secondary School certificate	Percentage: 95.8%

PROJECTS

UniVerse - CampusGPT Assistant for TIET Students



- Built a hybrid AI-powered campus assistant for Thapar University that answers real-time student queries on timetables, faculty, subjects, cafeteria menus, certificates, DoAA procedures, dispensary info, and campus navigation, using LLM-based intent detection integrated with structured (MySQL) and JSON-backed services.
- Implemented an interactive campus navigation system using a weighted graph of locations and paths, Dijkstra's shortest-path algorithm, Leaflet with OpenStreetMap for visualization, OSRM for routing, and distance-based walking time estimation.
- **Tech Stack** - MySQL, Google Gemini API, Vercel, Render, Python

Bangalore Traffic Prediction



- Built a Bangalore traffic congestion prediction system using historical traffic, weather, environmental impact, and temporal features, deployed via a FastAPI backend to deliver real-time congestion levels, categories, and analytical insights.
- Implemented a stacked Random Forest-XGBoost regression model with spatial and temporal feature engineering, achieving $RMSE = 2.031$ and $R^2 = 0.995$, outperforming baseline models and results reported in related research studies.
- **Tech Stack** - Python, FastAPI, Google Gemini API, HTML, CSS

CaloriesTracker



- Built a calorie tracking web application that allows users to upload food images and receive detailed insights on calories, nutrients, and health benefits using Google Gemini Pro Vision and LLM-based reasoning.
- Leveraged multimodal AI by combining image and text understanding to identify food items from images and generate natural language explanations on dietary value, health risks, and portion guidance.
- **Tech Stack** - Python, Generative AI, Streamlit

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Libraries & Frameworks: LangChain, Hugging Face Transformers, NumPy, Pandas, Scikit-learn, FastAPI

Artificial Intelligence/Machine Learning: Generative AI, Hugging Face, Retrieval Augmented Generation

Tools & Technologies: MySQL, Chroma, HTML/CSS, Ollama, Vercel, Linux, VS Code, Render

Core CS Concepts: Object-Oriented Programming (OOP), Operating Systems (OS), DBMS, Data Structures and Algorithms, Computer Networks, Artificial Intelligence

ACHIEVEMENTS

Gold medalist in 3600m cycle race.

Silver medalist in Skating Hockey (States)

CERTIFICATION

Software Engineer Intern (HackerRank)

Complete Data Science, Machine Learning, DL, NLP Bootcamp (IN-PROGRESS)